CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 4: Linear Inequations (In One Variable)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Solution set and the replacement set

1. Find the solution set of $0 < -\frac{x}{3} < 2, x \in Z$. (a) $\{-5, -4, -3, -2, -1\}$ (b) $\{-6, -5, -4, -3, -2, -1\}$ (c) $\{-5, -4, -3, -2, -1, 0\}$ (d) $\{-6, -5, -4, -3, -2, -1, 0\}$ ICSE 2025, 1 mark

SOLUTION

Multiply through by -3. Multiplying by a negative number reverses the signs: $0 > x > -6$.

The integers between -6 and 0 (both excluded) are $-5, -4, -3, -2, -1$.

Answer: (a)

2. Find the solution set of $2x + 4 \leq 14$, where x is a whole number. (a) $\{1, 2, 3, 4, 5\}$ (b) $\{0, 1, 2, 3, 4, 5\}$ (c) $\{1, 2, 3, 4\}$ (d) $\{0, 1, 2, 3, 4\}$ ICSE 2023, 1 mark

SOLUTION

$2x \leq 10$, so $x \leq 5$. Whole numbers start from 0.

Answer: (b)

2 Combining inequations

3. Solve $2x - \frac{5}{3} < \frac{3x}{5} + 10 \leq \frac{4x}{5} + 11, x \in R$. Write the solution set and show it on the number line. ICSE 2025, 3 marks

SOLUTION

Left part: $2x - \frac{3x}{5} < 10 + \frac{5}{3}$, so $\frac{7x}{5} < \frac{35}{3}$ and $x < \frac{25}{3}$.

Right part: $10 - 11 \leq \frac{4x}{5} - \frac{3x}{5}$, so $-1 \leq \frac{x}{5}$ and $x \geq -5$.

Solution set: $\left\{x : -5 \leq x < 8\frac{1}{3},\ x \in R\right\}$



Solid dot at −5 (included), hollow circle at 8⅓ (excluded)

4. Solve $-3 + x \leq \frac{7x}{2} + 2 < 8 + 2x$, where x is an integer. Write the solution set and show it on the number line. ICSE 2024, 3 marks

SOLUTION

Left part: $-3 - 2 \leq \frac{7x}{2} - x$, so $-5 \leq \frac{5x}{2}$ and $x \geq -2$.

Right part: $\frac{7x}{2} - 2x < 8 - 2$, so $\frac{3x}{2} < 6$ and $x < 4$.

Solution set: $\{-2, -1, 0, 1, 2, 3\}$



5. Solve $-5(x - 9) \geq 17 - 9x > x + 2, x \in R$. Write the solution set and show it on the number line. ICSE 2023, 3 marks

SOLUTION

Left part: $-5x + 45 \geq 17 - 9x$, so $4x \geq -28$ and $x \geq -7$.

Right part: $17 - 9x > x + 2$, so $15 > 10x$ and $x < 1.5$.

Solution set: $\{x : -7 \leq x < 1.5,\ x \in R\}$



6. Solve $\frac{3x}{5} + 2 < x + 4 \leq \frac{x}{2} + 5, x \in R$, and show the solution set on the number line. ICSE 2020, 3 marks

SOLUTION

Left part: $2 - 4 < x - \frac{3x}{5}$, so $-2 < \frac{2x}{5}$ and $x > -5$.

Right part: $x - \frac{x}{2} \leq 5 - 4$, so $\frac{x}{2} \leq 1$ and $x \leq 2$.

Solution set: $\{x : -5 < x \leq 2,\ x \in R\}$



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7. Solve $11x - 4 < 15x + 4 \leq 13x + 14$, where x is a whole number. Write the solution set and show it on the number line. ICSE 2019, 3 marks

SOLUTION

Left part: $-8 < 4x$, so $x > -2$.

Right part: $2x \leq 10$, so $x \leq 5$.

The whole numbers with $-2 < x \leq 5$ are 0, 1, 2, 3, 4, 5.

Solution set: $\{0, 1, 2, 3, 4, 5\}$



8. Solve $-2 + 10x \leq 13x + 10 < 24 + 10x$, where x is an integer. Write the solution set and show it on the number line. ICSE 2018, 3 marks

SOLUTION

Left part: $-12 \leq 3x$, so $x \geq -4$.

Right part: $3x < 14$, so $x < 4\frac{2}{3}$.

Solution set: $\{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$



9. Solve $-8\frac{1}{2} < -\frac{1}{2} - 4x \leq 7\frac{1}{2}$, where x is an integer, and show the solution set on the number line. ICSE 2017, 3 marks

SOLUTION

Add $\frac{1}{2}$ throughout: $-8 < -4x \leq 8$

Divide by -4 and reverse the signs: $2 > x \geq -2$

Solution set: $\{-2, -1, 0, 1\}$



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